

Simulation study

Methods

- 100 x 70 lattice with $KNN = 15$
- Purely inductive out of sample prediction
- Hold out sets have a completely independent graph
- 6 Fold CV:
 1. Block CV
 2. Random CV
- Simulated Autocorrelation:
 - X covariate with Moran's I of 0.379 and 0.686 respectively.
 - Errors at 0.390 and 0.770 respectively.
- Models used:
 - OLS
 - SEM (using GMM)
 - XGBoost
 - XGBoost + Spatial Lags
 - GraphSAGE
 - GrapheSAGE with node level LayerNorm
- Metric is $R^2 = 1 - SSE/SST$, averaged over six folds.

Scenario	Mean function	Error
A	Linear	Independent
B	Nonlinear	Independent
C	Linear	Spatially autocorrelated
D	Nonlinear	Spatially autocorrelated
E	Linear with a spatial lag in the response	Independent

Results

Spatially blocked folds:

Scenario	Graph-SAGE	GraphSAGE + LayerNorm	OLS	XG-Boost	XGBoost + lags	SEM
A	0.345	0.330	0.358	0.156	0.188	NA
B Ix=0.0	0.968	0.965	0.778	0.957	0.955	NA
B Ix=0.4	0.964	0.961	0.771	0.952	0.951	NA
B Ix=0.7	0.954	0.947	0.734	0.940	0.940	NA
C Iu=0.0	0.338	0.325	0.354	0.155	0.203	0.354
C Iu=0.4	0.299	0.297	0.342	0.144	0.157	0.343
C Iu=0.8	0.006	0.057	0.280	0.038	0.074	0.285
D Iu=0.0 Ix=0.4	0.937	0.936	0.455	0.912	0.914	0.455
D Iu=0.4 Ix=0.4	0.934	0.933	0.456	0.911	0.912	0.456
D Iu=0.8 Ix=0.7	0.928	0.921	0.386	0.905	0.904	0.398
E Ix=0.0	0.429	0.411	0.283	0.067	0.294	NA
E Ix=0.4	0.509	0.496	0.418	0.231	0.381	NA

Random folds:

Scenario	Graph-SAGE	GraphSAGE + LayerNorm	OLS	XG-Boost	XGBoost + lags	SEM
A	0.349	0.332	0.357	0.148	0.187	NA
B Ix=0.0	0.968	0.965	0.778	0.957	0.955	NA
B Ix=0.4	0.965	0.962	0.778	0.955	0.954	NA
B Ix=0.7	0.962	0.956	0.771	0.951	0.951	NA
C Iu=0.0	0.340	0.324	0.354	0.140	0.170	0.354
C Iu=0.4	0.315	0.317	0.356	0.146	0.175	0.356
C Iu=0.8	0.181	0.161	0.344	0.122	0.161	0.344
D Iu=0.0 Ix=0.4	0.938	0.937	0.456	0.916	0.916	0.456
D Iu=0.4 Ix=0.4	0.936	0.934	0.457	0.914	0.915	0.457
D Iu=0.8 Ix=0.7	0.934	0.932	0.429	0.914	0.914	0.426
E Ix=0.0	0.211	0.194	0.283	0.063	0.047	NA
E Ix=0.4	0.446	0.430	0.428	0.242	0.310	NA

- Spatial Lags almost always improve the XGBoost model
- Layer normalization didn't help except at high levels of autocorrelation in the residuals

Interpretation

- Nonlinearity (B, D) is where GraphSAGE Model improves the most
- Notably when there is a spatial lag in the response that where GraphSAGE works well as well (E).
- None of the DGPs are able to accurately observe the held out error so no amount of spatial lag in the covariates helps reconstructing.